

Section 6.5: Health and Safety

There are a number of **safety hazards** linked with using computers and ICT devices.

This section looks into some of these potential hazards and what you can do to prevent them.

We will also look at some of the **medical conditions** and **health problems** that prolonged use of ICT devices can cause and what you can do to avoid them.

Key Concepts of this section:

- # Understand the **health problems** related to the use of ICT equipment.
- # Know some strategies to **prevent these health problems**.
- # Be able to describe **safety issues** related to using computers and **measures for preventing accidents**.

Health problems

There are a number of health problems that you can suffer if you use **ICT devices incorrectly** or for **too long**. The main ones are:

- **Repetitive Strain Injury (RSI)**
- **Back and Neck Strain**
- **Eye Strain and Headaches**



Key Words:

RSI, Back Strain, Eye Strain, Headaches, Health Problems, Ergonomics

Description of these health problems, their causes and how to prevent them

1 Repetitive Strain Injury (RSI)

RSI causes **painful swelling of the wrist and fingers**. Sufferers with really bad RSI are unable to use their hands at all.

RSI is caused by **doing the same small movements over and over again** across a long period of time. **For example, clicking a mouse button repeatedly.**

Computer users who **type** and **use a mouse all day long** are commonly affected.

Causes of RSI

- Typing on a computer for too long.
- Using a mouse for long periods.
- Holding the mouse incorrectly.
- Working in a cramped workspace.

How to prevent RSI

- Take breaks to rest your hands.
- Use an ergonomic keyboard/mouse.
- Arrange your workspace so you are not cramped.
- Use a wrist rest.



Preventing RSI



Take breaks away from computer



Use an ergonomic keyboard



Spacious work area



Wrist rest

2 Back and Neck Problems

Back ache and neck ache can cause **great pain** and really affect the quality of your life.

Both back and neck ache can be caused by **sitting incorrectly** and using **poor quality chairs without back rests**. This is called **poor posture** (see below).

Causes of back and neck ache

- Working in a cramped workspace.
- Not sitting upright in your chair.
- Incorrect positioning of the computer screen.

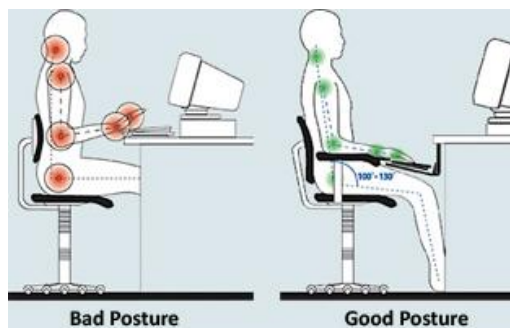
How to prevent back and neck ache

- Take regular breaks to stretch your body.
- Use adjustable chairs so you can sit in a position suitable for your height.
- Sit upright against the back rest.
- Tilt the computer screen so it is set just below your eye level.
- Keep your feet flat on the floor.



Seating Posture (click image to zoom)

The images to the right show you both the incorrect and correct ways to sit at a computer.



NOTE!

The key to reducing **wrist**, **finger**, **neck** and **back pains** is **ergonomics**.

Ergonomics focuses on creating **comfortable environments for people**. **Adjustable chairs**, **Wrist rests** etc are examples of ergonomics.

3 Eye Strain and Headaches

Staring at a computer screen for too long can **strain your eyes and cause headaches**.

Eye strain can cause your **vision to blur**.

Common causes of eye strain are [screen flicker](#) and having **direct light** causing [screen glare](#) (click links for video examples).

Causes of eye strain

- Staring at a computer screen for a long time.
- Working in a room with bad lighting.
- Using a computer screen with glare or flickers.
- Dirt on the screen.

How to prevent eye strain

- Take regular breaks (every hour or so).
- Use LCD screens rather than CRT as they have less flicker.
- Use an anti-glare screen.
- Ensure that room lighting is good with no direct light causing glare on the screen.
- Keep the screen clean of dirt.
- Have eyes tested regularly.



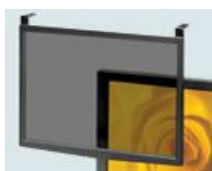
Preventing Eye Strain



Take breaks away from computer



Use an LCD screen rather than old CRT screen



Use an anti-glare screen



Keep the computer screen clean

Safety Issues

There are a number of safety issues that can arise from using ICT devices.

Some of these safety issues include:

- **Electrocution**
- **Tripping over wires**
- **Heavy equipment falling on you**
- **Fire risks**



Key Words:

Safety, Electrocution, Tripping, Heavy Equipment, Fire

Description of these safety issues

1 Electrocution

Most ICT devices require an **electrical power source**.

Whenever you have electrical power sources you run the **risk of the electrocution**.

Causes of Electrocution

- Faulty equipment (bare wires etc).
- Spilling drinks over electrical equipment.
- Opening up an electrical device when you don't know what you're doing.

How to Prevent Electrocution

- Make sure wires are insulated.
- Keep drinks away from equipment.
- Report any malfunctioning equipment to a technician.
- Never open up an electrical device.



Preventing Electrocution



Don't use exposed electrical wires



Keep drinks away from electrical equipment



Never open up an electrical device

2 Tripping over wires and cables

ICT devices with wires can make a room a potential hazard.

Trailing **wires are easy to trip over** if they are not secured or tucked away.

Causes of tripping over wires

- Long wires spread across a floor.

How to prevent tripping over wires

- Hide wires in cable ducts.
- Tuck trailing wires under desks or carpets.
- Use wireless technology to eliminate the use of wires altogether.



Preventing tripping over wires



Hide wires in cable ducts



Tuck wires under desks



Use wireless devices

3 Heavy equipment falling on you

Some ICT devices (like computer screens) are **fairly heavy** and can cause **injury if they fall on you**.

Equipment should be positioned **securely on strong desks** and tables well **away from the edge**.

Causes of equipment falling

- Equipment not positioned securely onto desks.
- Poor quality and flimsy desks.

How to prevent equipment falling

- Make sure that equipment is positioned away from the edge of desks.
- Use strong desks/tables that can support the weight with ease.



Preventing equipment falling off desks



Keep equipment away from the edge of desks



Use a strong desk that can take the weight

4 Fires

ICT devices require power from a mains outlet.

If **too many devices are plugged into a single mains at the same time** it is possible to overload the circuit and **start an electrical fire**.

You need to ensure that your room has many plug sockets so you don't have to overload any single one.

Causes of fires

- Too many devices plugged into a single mains outlet.
- Leaving devices plugged in unattended for long periods.
- Covering air vents on devices like laptops.

How to prevent fires

- Make sure that your room has plenty of mains outlets.
- Don't plug too many devices into the same outlet.
- Turn off and unplug devices if you are going to be away for a long time.
- Have a CO2 fire extinguisher at hand.
- Leave air vents on devices uncovered.



Preventing electrical fires



Don't overload mains outlets (plug sockets)



Have a fire extinguisher in the room



Turn off and unplug devices when not in use



Do not cover up air vents on devices

Activities!

[Health and Safety - Questions](#)

Click the above task and follow the instructions given on the sheet. Make sure that you complete all questions in full.



Click on the image to the left and see how many of the potential hazards you can spot



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[Section 7: The Ways in which ICT is used](#)

[Section 2: Input and Output Devices](#)

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[Section 7: Part 2](#)

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